

RESEARCH REPORT

Kolmogorov-Arnold Networks for Single-Molecule Localization Microscopy (KAN-SMLM)

1. Research Overview

University	City College of New York
Department	Electrical Engineering
Degree Program	Doctor of Philosophy (PhD)
Advisor	Professor Yi Sun
Report Period	August 2025 - June 2026
Primary Research Domain	KAN for single-molecule localization microscopy (SMLM)

Core Research Question

Can Kolmogorov-Arnold Networks (KANs), whose learnable edge functions are based on the Kolmogorov-Arnold representation theorem, be developed into a practical and interpretable architecture for detecting and localizing emitters in SMLM?

2. KAN Literature Reviewed and Used to Guide the Research

The papers below were used to understand the KAN theory, implementation choices, alternative basis functions, and image-model extensions that are most relevant to KAN-SMLM. The list includes the foundational KAN work as well as papers that helped frame the Gaussian/Mexican-hat basis functions choices, the pure KAN experiments, and the later CNN+KAN direction.

Reference	Title / Description	Why It Matters for my research
Kolmogorov, 1957	On the representation of continuous functions of many variables by superposition of continuous functions of one variable and addition	Mathematical foundation for the representation theorem behind KAN.
Liu et al., 2024	KAN: Kolmogorov-Arnold Networks	Main KAN paper. Introduces learnable univariate edge functions and spline-based parameterization.
Liu et al., 2024	KAN 2.0: Kolmogorov-Arnold Networks Meet Science	Useful for the interpretability and scientific-machine-learning motivation of KANs.
Github 2024	Efficient-KAN: An Efficient Implementation of Kolmogorov-Arnold Network	Practical vectorized PyTorch implementation of B-spline KAN layers; relevant to the initial pyKAN/implementation comparison.
Li, 2024	Kolmogorov-Arnold Networks are Radial Basis Function Networks (FastKAN)	Motivates testing Gaussian/RBF-style functions as computationally simpler alternatives to cubic B-splines.
Qiu et al., 2024	ReLU-KAN: New Kolmogorov-Arnold Networks that Only Need Matrix Addition, Dot Multiplication, and ReLU	Shows one approach to reducing B-spline computational cost using ReLU-based basis functions.
Bozorgasl & Chen, 2024	Wav-KAN: Wavelet Kolmogorov-Arnold Networks	Relevant for using multi-scale functions, including Mexican-hat-like wavelet ideas, in a KAN layer.
Sidharth et al., 2024	Chebyshev Polynomial-Based Kolmogorov-Arnold Networks	Alternative polynomial basis for nonlinear function approximation and computational efficiency.
Ta, 2024	BSRBF-KAN: B-splines and Radial Basis Functions in KANs	Related to comparing pure B-splines with Gaussian/RBF basis functions and hybrid combinations.

Reference	Title / Description	Why It Matters for my research
Bodner et al., 2024	Convolutional Kolmogorov-Arnold Networks (CKAN)	Instead of applying fixed non-linearities (like ReLU) after a linear dot product, CKANs place learnable univariate functions on the edges (kernels).
Drokin, 2024	Kolmogorov-Arnold Convolutions: Design Principles and Empirical Studies	Provides design alternatives and empirical discussion for KAN-based convolutional layers in vision tasks.
Cheon et al., 2024	Demonstrating the Efficacy of Kolmogorov-Arnold Networks for Vision	Useful reference for the practical limitations and design requirements of KANs.
Patra et al., 2024	Physics-Informed Kolmogorov-Arnold Neural Networks for Dynamical Analysis	Supports the physics-informed KAN direction and motivates encoding known system structure rather than learning everything from raw inputs.
Jacob et al., 2024	Separable Physics-Informed Kolmogorov-Arnold Networks	Relevant to physics-informed KAN design and to the computational challenge of high-dimensional inputs.
Hou et al., 2024	Kolmogorov-Arnold Networks: A Critical Assessment	Important for keeping the comparison cautious: KAN benefits should be tested under fair settings, especially for noisy and high-dimensional data.
Somvanshi et al., 2024	A Survey on Kolmogorov-Arnold Network	Broad map of KAN variants, applications, efficiency issues, and hybrid architectures.

3. Research Timeline

The dates below are organized as approximate working periods.

Approx. Period	Research Task	Technical Details and Main Inference
Aug-Sep 2025	KAN theory and initial literature review	I started by reading the original KAN paper and related work to understand the Kolmogorov-Arnold representation theorem, learnable edge functions, cubic B-splines, spline grids, and why KANs are different from ordinary MLPs. I also looked at the possible use of KAN for a physics-based imaging problem such as SMLM.
Sep-Oct 2025	pyKAN testing on the 64 x 64 circle-emitter dataset (dataset professor created and published on github)	I used pyKAN on the synthetic 64 x 64 circle-emitter dataset available in the lab GitHub repository, where each frame can contain a different number of emitters. I first used a sliding-window/cropping approach. The KAN estimated the probability that an emitter was present in each window and then localized the emitter inside the selected window. This gave a strong initial result. When I removed the crop and gave the full image directly to the KAN, performance dropped. This suggested that the window was helping by separating detection from fine localization.
Oct-Nov 2025	Basis-function comparison: B-spline versus Gaussian and Mexican-hat functions	I changed the edge-function basis to test Gaussian/RBF-like and Mexican-hat functions, both as replacements for and in combination with B-splines. This was motivated by the Gaussian shape of the SMLM point-spread function and by the possibility that Mexican-hat functions could help separate a local peak from background. The comparison showed that the pure B-spline configuration worked better for this task than the tested alternatives or hybrids.
Dec 2025-Jan 2026	Reading CNN+KAN and CKAN literature	I read CNN+KAN and convolutional KAN (CKAN) papers to understand how convolutional feature extraction can be combined with KAN nonlinear function learning for image data. At this point, I did not run a CNN+KAN or CKAN algorithm; this stage was mainly for learning the architecture options and deciding what should be tested later.
Dec 2025-Jan 2026	Controlled emitter datasets at 32 x 32 and 16 x 16 pixels	I created simplified datasets with 1 to 3 emitters per frame at 32 x 32 and 16 x 16 pixel resolutions. Running KAN on these controlled datasets gave good results. This was useful because it confirmed that the basic KAN setup could learn the localization problem when the input size and problem complexity were reduced.
Jan-Feb 2026	Custom KAN architecture instead of relying on pyKAN	Because pyKAN was slow and because I wanted a more novel and controllable implementation, I developed my own KAN architecture and code. I tested it on the 32 x 32 and 16 x 16 multi-emitter datasets. The first custom implementation did not give good results, which showed that reproducing the basic KAN behavior, especially the spline implementation and training stability, was not straightforward.

Approx. Period	Research Task	Technical Details and Main Inference
Jan-Feb 2026	Preprocessing experiments	I tested a preprocessing pipeline before feeding the images to the KAN, including local background subtraction and intensity normalization. This improved performance. However, Professor noted that the main comparison should avoid relying on preprocessing and should evaluate whether a pure KAN can learn directly from raw frames.
Jan-Feb 2026	Physics-informed feature input	Instead of passing only raw pixel values, I constructed an eight-feature physics-informed input vector based on expected SMLM/PSF properties: total signal, peak intensity, estimated local background, signal-to-background ratio, intensity-weighted x centroid, intensity-weighted y centroid, x-direction spread, and y-direction spread. Feeding these features to the network gave good results, showing that known PSF and photon-distribution information can make the learning problem easier. The trade-off is that this is no longer a pure raw-pixel KAN.
March-Apr-May 2026	Pure KAN on raw 7 x 7 patches: 21 configurations	I returned to a pure KAN setup using raw, unprocessed 7 x 7 image patches. I tested 21 versions by changing architectural and training parameters such as network width/depth, spline-grid settings, learning rate, batch size, and training duration. I began with random emitter locations (M=1-5) and (M=0 to 8) and then repeated the tests with fixed emitter locations to reduce position variability and identify whether the remaining difficulty was primarily caused by the model or by the location distribution. This stage produced acceptable and then good pure-KAN results without preprocessing.
May 2026	CNN+KAN experiments: six configurations	I implemented and tested six CNN+KAN versions. The CNN part was used to learn local spatial features from the dataframe patch, while the KAN part was used as the final nonlinear regression head for localization. The best CNN+KAN version gave the strongest overall results on both fixed-location and random-location emitter datasets.

4. Programming Languages, Tools, and Environment

Platform / Language	Libraries or Tools	Used For
Python	PyTorch, NumPy, SciPy, Pandas, Matplotlib, PIL, tiffle	Dataset loading, model implementation, training, evaluation, and visualization.
Python	pyKAN	Initial KAN experiments and reference implementation before the custom KAN attempt.
Python	Custom KAN code	Testing a faster and more controllable B-spline KAN implementation for the 32 x 32, 16 x 16, and 7 x 7 datasets.
MATLAB / Python	Synthetic emitter-data generation	
Google Colab	GPU runtime and Drive mounting	Primary training and testing environment.
VS Code / Cursor	Local coding and debugging	Code development and experiment management.